

DBCLS BioHackathon 2026 report: QPX: Pathway analysis environment for non-model organisms

BioHackathon series:
DBCLS BioHackathon 2026
Matsuyama, Japan, 2026
QPX

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Introduction

As part of the DBCLS BioHackathon 2026, we here report the development of pathway analysis environment called Quest for Pathways with eXpression (QPX) designed for non-model organisms.

Meeting information

If you want to submit a preprint to BioHackrXiv, first check if your meeting is registered. You can find a list of meetings [here](#). If your meeting is missing, please contact your meeting organizers. The above list also provides information on the YAML fields with information about the meeting.

The following fields need to be given:

```
biohackathon_name: "DBCLS BioHackathon 2026"
biohackathon_url:  "https://2026.biohackathon.org/"
biohackathon_location: "Matsuyama, Japan, 2026"
group: YOUR-PROJECT-NAME-GOES-HERE
git_url: https://github.com/yourOrganization/your_report_repo
```

The [BioHackrXiv meeting pages](#) provide content to use for the first three fields. The `git_url` field must have the link to the GitHub repository with your preprint (draft).

Author information

Information about the authors is given in the [YAML](#) format at the top of this template. For authors you provide their names, their affiliations. That is the minimum, but as BioHackrXiv is moving to a situation where more metadata is shared, and used by, for example, EuropePMC, adding additional information is encouraged.

BioHackathons is about hacking together, and the minimal number of authors for reports is two. This makes a minimal example look like this:

```
authors:
  - name: First Author
    affiliation: 1
  - name: Last Author
    affiliation: 2
affiliations:
  - name: First Affiliation
    index: 1
  - name: ELIXIR Europe
    index: 2
```

Author identifiers

Ideally, authors provide their [ORCID](#) identifier. For affiliations, It is added with the `orcid:` field. So, an author record would look like this:

```
authors:
  - name: First Author
    affiliation: 1
    orcid: 0000-0000-0000-0000
```

Research Organization Registry identifiers

Matching the author identifier, the affiliations can be further specified with the [Research Organization Registry](#) (ROR) identifier. For example, this is the affiliation identifier can be added with the `ror:` field:

```
affiliations:
  - name: ELIXIR Europe
    ror: 044rwnt51
    index: 2
```

Contributor Role Taxonomy

A last feature since is minimal support for the Contributor Role Taxonomy (CRediT). You can specify the role of authors in writing the report with the `role:` field. However, the authors are responsible for selection the right terms from [CRediT](#). An example looks like this:

```
authors:
  - name: First Author
    affiliation: 1
    orcid: 0000-0000-0000-0000
    role: Conceptualization, Writing - review & editing
```

A full examples

A full example then has this structure:

```
authors:
  - name: First Author
    affiliation: 1
    role: Writing - original draft
  - name: Last Author
    orcid: 0000-0000-0000-0000
    affiliation: 2
    role: Conceptualization, Writing - review & editing
affiliations:
  - name: First Affiliation
    index: 1
  - name: ELIXIR Europe
    ror: 044rwnt51
    index: 2
```

Formatting

This document use Markdown and you can look at [this tutorial](#).



Subsection level 2

Please keep sections to a maximum of only two levels.

Tables

Tables can be added in the following way, though alternatives are possible:

Table: Note that table caption is automatically numbered and should be given before the table itself.

```
| Header 1 | Header 2 |
| ----- | - - - - - |
| item 1  | item 2  |
| item 3  | item 4  |
```

This gives:

Table 1: Note that table caption is automatically numbered and should be given before the table itself.

Header 1	Header 2
item 1	item 2
item 3	item 4

Figures

A figure is added with:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png)
```

This gives:



Figure 1: Caption for BioHackrXiv logo figure

Figures can be scaled by adding the width or height to the Markdown like this:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png){ width=50px }
```

You can add cross references to figures by adding a LaTeX `\label{figureCode}` to the label of the Markdown figure and then use `\ref{figureCode}` to cite it:

```
![Caption for BioHackrXiv logo figure \label{figureCode}](./biohackrxiv.png){ width=50px }
```

This way, we can cite Figure [1](#).

Other main section on your manuscript level 1

Lists can be added with:

1. Item 1
2. Item 2

Citation Typing Ontology annotation

You can use [CiTO](#) annotations, as explained in [this BioHackathon Europe 2021 write up](#) and [this CiTO Pilot](#). Using this template, you can cite an article and indicate *why* you cite that article, for instance DisGeNET-RDF (Queralt-Rosinach et al., [2016](#)).

The syntax in Markdown is as follows: a single intention annotation looks like `[@usesMethodIn:Krewinkel2017]`; two or more intentions are separated with colons, like `[@extends:discusses:Nielsen2017Scholia]`. When you cite two different articles, you use this syntax: `[@citesAsDataSource:Ammar2022ETL; @citesAsDataSource:Arend2022BioHackEU22]`.

Possible CiTO typing annotation include:

- `citesAsDataSource`: when you point the reader to a source of data which may explain a claim
- `usesDataFrom`: when you reuse somehow (and elaborate on) the data in the cited entity
- `usesMethodIn`
- `citesAsAuthority`
- `citesAsEvidence`
- `citesAsPotentialSolution`
- `citesAsRecommendedReading`
- `citesAsRelated`
- `citesAsSourceDocument`
- `citesForInformation`
- `confirms`
- `documents`
- `providesDataFor`
- `obtainsSupportFrom`
- `discusses`
- `extends`
- `agreesWith`
- `disagreesWith`
- `updates`

There is a general `cites` intention, but this is already implied and should be left out.

Results

Discussion

...

Acknowledgements

...

References

Author contributions

Hidemasa Bono: Conceptualization, Writing - original draft; Naoya Oec: System development, Validation; Haruka Tanimoto: Data creation; Tomohiro Taguchi: Data creation; Amane Kurata: Data creation; Ryo Nozu: Validation, Data creation.

References

Queralt-Rosinach, N., Pinero, J., Bravo, À., Sanz, F., & Furlong, L. I. (2016). DisGeNET-RDF: harnessing the innovative power of the Semantic Web to explore the genetic basis of diseases. *Bioinformatics*, 32(14), 2236–2238. <https://doi.org/10.1093/bioinformatics/btw214> [cito:citesAsAuthority].



Appendices

If you want the Appendix (-ces) to show up after the references, wrap them in after the header, like done in this Markdown file. Look at the [source](#) to see the exact structure.